

HW 10

Stochastic Process

Problem 14.2.

A company's cash position, measured in millions of dollars, follows a generalized Wiener process with a drift rate of 0.5 per quarter and a variance rate of 4.0 per quarter. How high does the company's initial cash position have to be for the company to have a less than 5% chance of a negative cash position by the end of one year?

Problem 14.4.

Consider a variable, S , that follows the process

$$dS = \mu dt + \sigma dz$$

For the first three years, $\mu = 2$ and $\sigma = 3$; for the next three years, $\mu = 3$ and $\sigma = 4$. If the initial value of the variable is 5, what is the probability distribution of the value of the variable at the end of year six?

Problem 14.8.

It has been suggested that the short-term interest rate, r , follows the stochastic process

$$dr = a(b - r)dt + rc dz$$

where a , b , and c are positive constants and dz is a Wiener process. Describe the nature of this process.

Problem 14.10.

Suppose that x is the yield to maturity with continuous compounding on a zero-coupon bond that pays off \$1 at time T . Assume that x follows the process

$$dx = a(x_0 - x)dt + sx dz$$

where a , x_0 , and s are positive constants and dz is a Wiener process. What is the process followed by the bond price?

Problem 14.12.

Suppose that a stock price has an expected return of 16% per annum and a volatility of 30% per annum. When the stock price at the end of a certain day is \$50, calculate the following:

- The expected stock price at the end of the next day.
- The standard deviation of the stock price at the end of the next day.
- The 95% confidence limits for the stock price at the end of the next day.